Linear Programming

ivation: What is O

Generally, we can write the formulation as

minimize $c_1 x_1 + c_2 x_2 + \cdots + c_n x_n$

subject to $a_{11} x_1 + a_{12} x_2 + \cdots + a_{1n} x_n \geq b_1$

$a_{21} x_1 + a_{22} x_2 + \cdots + a_{2n} x_n \geq b_2$

$\vdots$

$a_{m1} x_1 + a_{m2} x_2 + \cdots + a_{mn} x_n \geq b_m$

$x_1, x_2, \ldots, x_n \geq 0$

$\Leftrightarrow$

minimize $\begin{bmatrix} c_1 & c_2 & \cdots & c_n \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$

subject to: $\begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} \geq \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{bmatrix}$

$x_1, x_2, \ldots, x_n \geq 0$

Prototypical example: the Diet Problem

We have a limited budget and

※ Standard form Problems

A linear programming problem of the form

$$\begin{aligned} &\text{minimize} && c^T x \\ &\text{subject to} && Ax = b \\ & && x \geq 0 \end{aligned}$$

is said to be in standard form.

※ Reduction to Standard form

a.) Elimination of free variables: Given an unrestricted variable x_j, we replace it by $x_j^+ - x_j^-$, where x_j^+, x_j^-

b.) Elimination of Inequality constraints: Given

$$\sum_{j=1}^{n} a_{ij} x_j \leq b_i$$

introduce a slack variable $s_i \geq 0$ so that we have

$$\sum_{j=1}^{n} a_{ij} x_j + s_i = b_i$$

Similarly for $\sum_{j=1}^{n} a_{ij} x_j \geq b_i$, introduce a surplus variable $s_i \geq 0$, so that we have

$$\sum_{j=1}^{n} a_{ij} x_j - s_i = b_i$$

Remark: $\max \{ c^T x \mid Ax \; b, x \geq 0 \} = -\min \{ -c^T x \mid Ax \leq b, \; x \geq 0 \}$.

Negating a set of numbers flips its ordering. For any function $f(x)$ over a feasible set S:

$$\sup_{x \in S} f(x) = -\inf_{x \in S} \left(-f(x) \right)$$

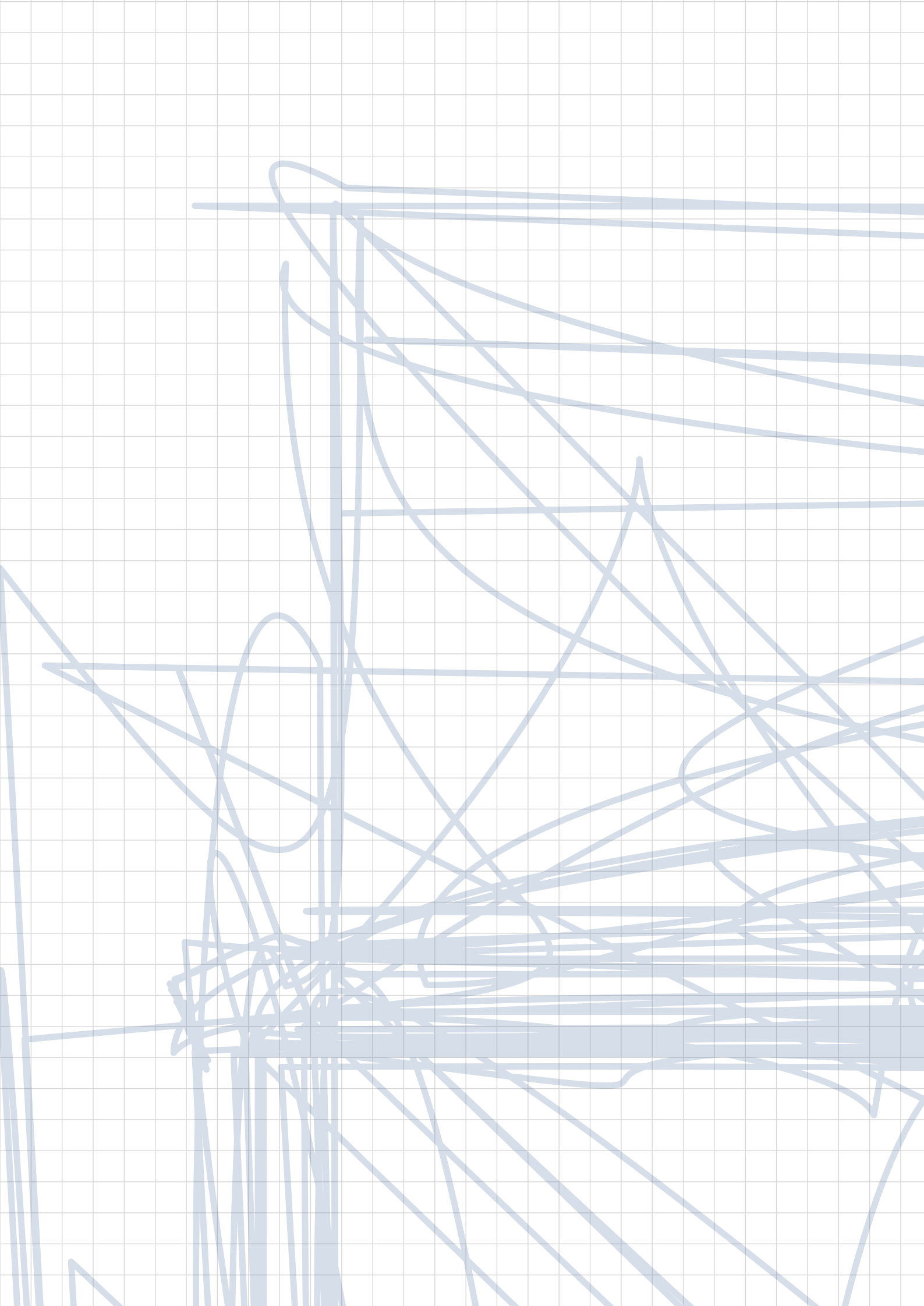
If the problem has an achievable optima, this is written as

$$\max_{x \in}$$

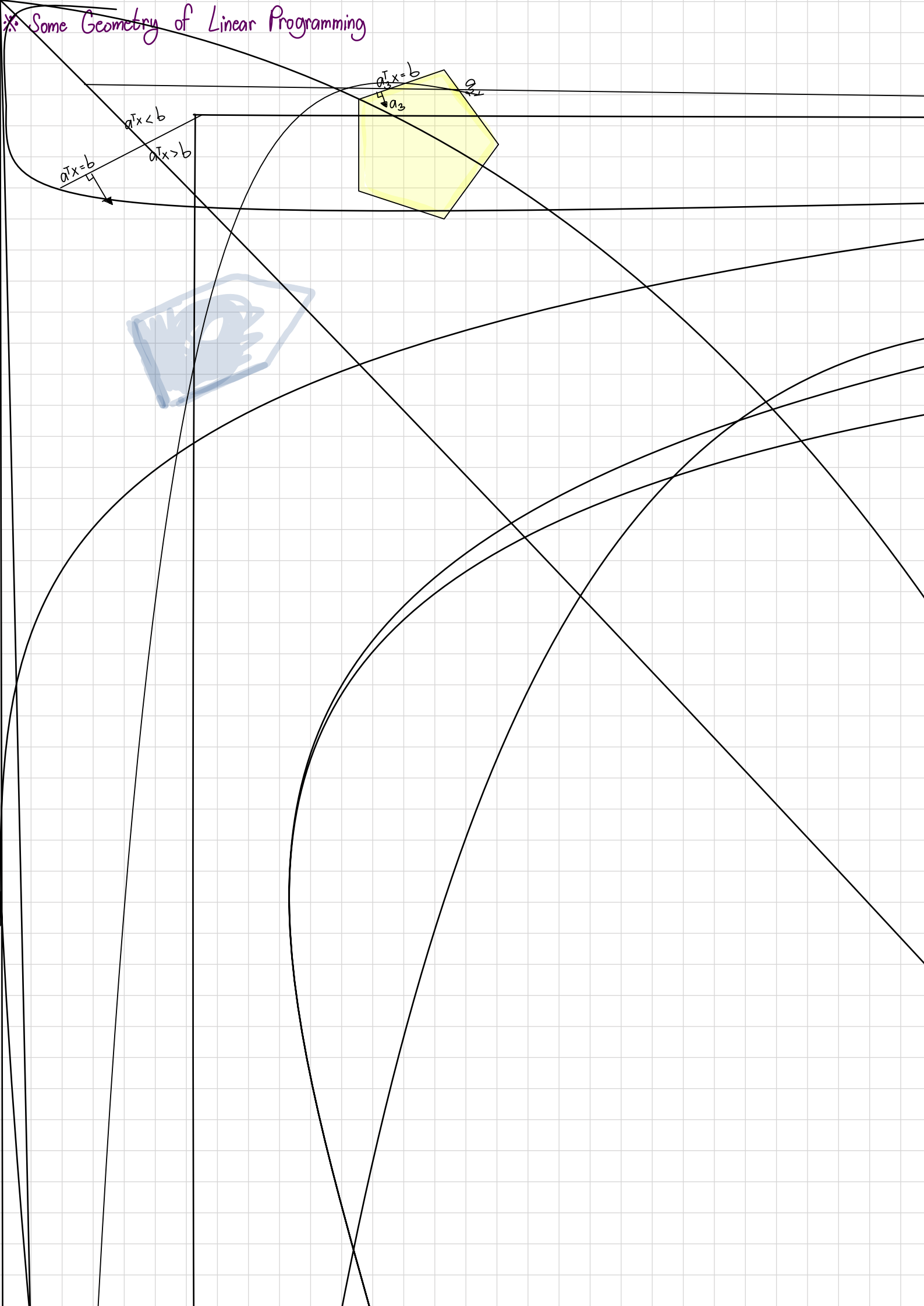
※ Resource - Allocation Problem : Wyndor Glass Co.

- Plant 1 → produces Aluminum frames
- Plant 2 → produces wood frames
- Plant 3 → produces glass and assembles the products

  - Product 1 : An 8-foot glass door aluminum framing → Plants 1 & 3
  - Product 2 : A 4 x 6 foot double-hung wood-framed window →



Some Geometry of Linear Programming



If x ... basic solution then

- $x_{B(1)}$, ... $x_{B(m)}$ are called basic variables; the ... g
- $A_{B(1)}$, ... $A_{B(m)}$ are called basic columns

3.) Compute $u = B^{-1}A_j$. If no component u is positive, we have $\theta^* = \infty$, the optimal cost is $-\infty$, and the algorithm terminates.

$$u = B^{-1}A_2 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ 2 \\ 2 \end{bmatrix} = \begin{bmatrix} 0 \\ 2 \\ 2 \end{bmatrix} \begin{matrix} \rightarrow u_{B(1)} \\ \rightarrow u_{B(2)} \\ \rightarrow u_{B(3)} \end{matrix}$$

4.) If some component u is positive, let

$$\lambda^* = \min_{\{i=1,\dots,m \mid u_i > 0\}} \frac{x_{B(i)}}{u_i}$$

$$\lambda^* = \min\left\{\frac{x_{B(2)}}{u_2}, \frac{x_{B(3)}}{u_3}\right\} = \min\left\{\frac{12}{2}, \frac{18}{2}\right\} = \min\left\{\frac{12}{2}, \frac{18}{2}\right\} = \min\{6, 9\} = 6$$

5.) Let l be such that $\lambda^* = \frac{x_{B(l)}}{u_l}$. Form a new basis by replacing